BADRUN NAHAR LUNA

Software Engineer & Aspiring AI Engineer

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CAREER OBJECTIVE

Motivated Computer Science graduate with expertise in Machine Learning, Deep Learning, and data-driven research. Experienced in academic research, intelligent model development, and teaching technical concepts. Proficient in Python with additional skills in HTML and CSS. Seeking opportunities as a Research Assistant or Software Engineer to apply research and programming skills to develop impactful technological solutions.

PROFESSIONAL EXPERIENCE

Trainee Software Engineer - (Intern) **Feb 10 2026 - Present**

Bangladesh Software Solutions

- Assist in software development, testing, and debugging under supervision.
- Collaborate with the team on projects and tasks as assigned.
- Participate in training sessions and skill development programs.

Research Assistant (Part Time) **July 2025 - Present**

InsightEdu T&RS *Team Lead*

- Conduct literature reviews and contribute to writing and editing research papers for academic publication.
- Perform data cleaning, preprocessing, and exploratory data analysis to ensure high-quality datasets.
- Design, train, and evaluate machine learning and deep learning models tailored to project objectives.
- Collaborate with senior researchers to optimize methodologies and validate experimental results.

Teaching Assistant (TA) **Jan 2025 - Dec 2025**

Bangladesh University of Business and Technology (BUBT)

- Conducting lab sessions and tutorials on Java and Object-Oriented Programming (OOP).
- Previously assisted in Microprocessor and Microcontroller courses, providing technical support in circuit design and embedded systems programming.
- Mentored students' technical projects and debugged their code.
- Collaborated with faculty to enhance lab materials and student engagement.

EDUCATION

Bachelor of Science in Computer Science & Engineering **July 2021 - June 2025**

Bangladesh University of Business and Technology (BUBT)

CGPA 3.93/4.00

- Major in Artificial Intelligence (AI).
- Thesis on "Efficient and Scalable Bird Species Classification Using an Ensemble of Deep Learning Models".

Higher Secondary Certificate (HSC) **June 2018 - April 2020**

Viqarunnisa Noon School and College

GPA: 5.00 out of 5.00

Secondary School Certificate (SSC) **Jan 2008 - Feb 2018**

Viqarunnisa Noon School and College

GPA: 5.00 out of 5.00

ACHIEVEMENTS AND AWARDS

- **Second Position,** Project Showcasing – Brainstorming Week 2024 (BUBT)
- **Champion,** HULT Prize BUBT Campus Round (2022 & 2023)
- **Champion,** Science Fest – Cantonment Public School and College (2019)
- **Champion,** Science Festival - Shaheed Bir Uttam Anwar Girls' School & College (2017)

RESEARCH & ACADEMIC PROJECTS

Bird Species Classification Using Ensemble Deep Learning and Vision Transformers

- ***Accepted*** *as a conference paper to the 2025 28th International Conference on Computer and Information Technology (ICCIT)*

Built an advanced bird classification model using EfficientNet, ResNet50, InceptionResNetV2, and Vision Transformer (ViT-B32). Applied ensemble learning and fine-tuning to boost accuracy. The project demonstrates expertise in transfer learning, transformer models, and image classification.

A Hybrid CNN-Transformer Framework with Generative Balancing for Dermoscopic Image Classification

- ***Accepted*** *as a conference paper to the 2025 11th IIEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*

Developed a Hybrid Convolutional Transformer Network for automated skin lesion classification using the ISIC-2019 dataset. Incorporated a Conditional GAN to address class imbalance by synthesizing minority-class dermoscopic images. Leveraged EfficientNetB3 for local feature extraction and DeiT-S for global feature encoding, combined with a focal loss-based classification head optimized via AdamW. Demonstrate a robust state-of-the-art solution for skin cancer detection.

Autonomous Healthcare Monitoring via IoT-Edge Computing

- ***Accepted*** *as a conference paper to the International Symposium on Biomedical Monitoring (ISBM 2025)*

Developed a real-time health monitoring solution using ESP8266, MAX30100, DS18B20, Servo Motor, and SMPS. The system measures heart rate, body temperature, and oxygen saturation, triggering automated alerts when abnormal readings are detected. The project showcased the integration of IoT and health informatics for patient safety.

A Time-Grounded Bengali Political Sentiment Dataset for Temporal Visual Analytics of Fine-Grained Political Sentiment Evolution

- ***Accepted*** *as a B* conference paper to the VA4MDS 2026 Workshop (IEEE PacificVis 2026)*

Developed a time-grounded Bengali political sentiment analysis framework using a 3,300-sample context-aware dataset aligned with major sociopolitical events. A lightweight BanglaBERT model with an attention-refinement mechanism was proposed to capture temporal sentiment dynamics, achieving 93.52% accuracy and 92.70% macro-F1 with low computational complexity.

DenseUNet-SC: A Multi-Task Deep Learning Framework for Brain Tumor Segmentation and Classification

Developed a unified deep learning model for brain tumor segmentation and classification using MRI images to support clinical decision-making. The framework combines both tasks in a single architecture, enabling shared feature learning and improved diagnostic consistency. By leveraging segmentation information to guide classification, the model enhances robustness across different MRI scans. Experimental evaluation on a public dataset demonstrated high classification accuracy and reliable tumor segmentation performance.

RESEARCH INTERESTS

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Regression and Classification Models
- Image Processing
- Deep Learning (DL)
- Explainable AI (XAI)
- Computer Vision
- Convolutional Neural Network (CNN)

PROFESSIONAL TRAINING

QC Analyst – Job Ready Program **April 2025**

FastFlowUp & PTE Platform Duration : 5 Days

- Hands-on training in AI-driven software testing and quality control.
- Worked as a paid QC Analyst on a live PTE platform.

TECHNICAL PROFICIENCIES

- **Languages:** Python (Advanced), Java, C, C++
- **Tools:** Microsoft Office, PowerPoint, Excel, Word, Git, GitHub
- **Database:** MySQL, MongoDB, Firebase
- **Software:** HTML, CSS
- **Technologies:** Machine Learning, Deep Learning, Computer Vision, Prompt Engineering, Blynk, Cloud Deployment (Firebase, Hugging Face Spaces)
- **Frameworks:** TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, NLTK, Pandas, NumPy
- **Model Development:** Dataset Preparation, Data Collection, EDA, Data Preprocessing, Embedding Generation, Model Training, Fine-tuning, Model Evaluation
- **Other Skills:** Video Editing, Graphics Design, Documentation, Presentation Design

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Machine Learning – *kaggle*

- Completed a hands-on course covering supervised and unsupervised learning, model evaluation, and feature engineering using real-world datasets.

Natural Language Processing for Generative AI – *Innovative Skills*

- Completed practical projects involving tokenization, embeddings, sentiment analysis, and transformer-based models like BERT.

Research Learning Program 2024 – *Bangladesh Research Society (BDRS)*

- Completed the six-month course enhanced my skills in academic research, critical thinking, and collaborative inquiry.

English Communication Skills – *Mentors*

- Completed a structured English course focused on academic, professional, and conversational fluency through guided mentorship.

LEADERSHIP & EXTRACURRICULAR ROLES

- **Chairperson,** IEEE WIE BUBT Student Branch Affinity Group (2022–June 2025)
- **Vice-Chair,** IEEE BUBT Student Branch (2021– June 2025)
- **Co-Convener,** BASIS Students' Forum of BUBT Chapter (2022– June 2025)
- **Volunteer,** ICPC Dhaka Regional Contest (2022, 2023)

REFERENCE

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